

[JPJC/2025/Q1]

- 1 You are designing a simple text-based treasure-hunting game. After each game session, the player collects treasures. Each treasure has:
 - A name (string)
 - A value in gold coins (integer)

You need to implement a system that allows the player to sort and search their treasure inventory.

Task 1.1

Write a function `readData(filename)` that takes a string `filename` which represents the name of a csv file, reads in the contents of the file, and returns the content.

For example:

```
[["Sapphire Crown", 230], ["Rusty Dagger", 75], ["Dragon Scale", 480]]
```

Call your function `readData` with the file `TASK1.csv` and display the returned list. [5]

Task 1.2

Write a function `quicksort(treasures)` that takes the returned list from **Task 1.1**, implements a **recursive** quicksort algorithm (sorted by its treasure name in ascending order lexicographically, and returns the sorted list.

For example:

```
[["Dragon Scale", 480]], ["Rusty Dagger", 75], ["Sapphire Crown", 230]]
```

Call your function `quicksort` with the list from **Task 1.1** and display the returned ordered list. [6]

Task 1.3

Write a function `binarySearch(treasures, name)` that takes the returned list from **Task 1.2**, implements an **iterative** binary search on the name. If the name exists, it returns the value of the treasure. Otherwise, it returns `-1`.

Call your function `binarySearch` with the list from **Task 1.2** to search for "Dragon Scale" and display the returned value. [3]

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Task 1.4

Write a Python program that will:

- load the data from file `TASK1.csv`
- display the menu and prompt for user choice
- execute the option based on user choice by calling the appropriate functions defined in **Tasks 1.1 to 1.3**
- repeat the above steps to display the menu and execute user choice
- exit the program when user enters option 4.

For “Search Treasure by Name” option, if the treasure name cannot be found, display the message “Treasure cannot be found”.

Sample Output:

```
Welcome to Your Treasure Inventory!
1. View Inventory
2. Sort Treasures by Name
3. Search Treasure by Name
4. Exit

Choose option: 1
Sapphire Crown (Value: 230)
Rusty Dagger (Value: 75)
Dragon Scale (Value: 480)

Welcome to Your Treasure Inventory!
1. View Inventory
2. Sort Treasures by Name
3. Search Treasure by Name
4. Exit

Choose option: 2
Dragon Scale (Value: 480)
Rusty Dagger (Value: 75)
Sapphire Crown (Value: 230)

Welcome to Your Treasure Inventory!
1. View Inventory
2. Sort Treasures by Name
3. Search Treasure by Name
4. Exit

Choose option: 3
Enter treasure name to search: Dragon Scale
Found 'Dragon Scale' worth 480 gold coins!

Welcome to Your Treasure Inventory!
1. View Inventory
2. Sort Treasures by Name
3. Search Treasure by Name
4. Exit

Choose option: 4
Goodbye!
```

Run your program to test all the four menu options and display the output.

Save your Jupyter Notebook for Task 1.

[6]